



WAN Automation Framework

Infrastructure Resiliency & Deployment Orchestration

TechLink Solutions

High Availability – Reliability – Fault Tolerance – Resiliency - Scalability

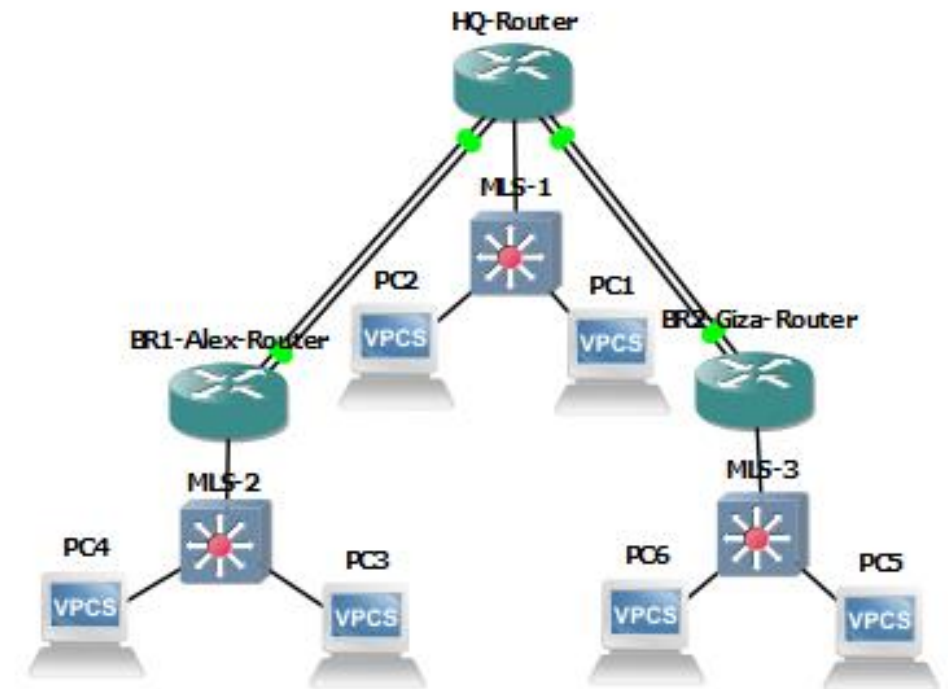
NETWORK TOPOLOGY OVERVIEW

Architecture & Topology Resilience

Strategic multi-branch footprint linking Cairo HQ core to Alexandria and Giza regional nodes.

- **Dual WAN Ingress:** Redundant ISP paths per branch terminating on Core L3.
- **Performance & Efficiency:** Optimizing traffic paths and localized routing decisions to preserve expensive WAN bandwidth and keep latency to an absolute minimum.
- **High Availability:** Physical diversity across Gi1/0 to Gi4/0 interfaces.

Logical Segmentation: Strict separation of the management plane (VLAN 10) from end-user traffic (VLAN 20) across all local access ports



DYNAMIC ROUTING ENGINE



OSPF Area 0

Single Backbone Area 0
configuration for maximum
convergence speed and simplicity.



Traffic Flow

Sub-second RIB recalculation upon
link state failure ensures zero-
touch recovery.



Why OSPF

dynamic neighbor discovery and
zero manual route management

The Failover Mechanism

Link Failure

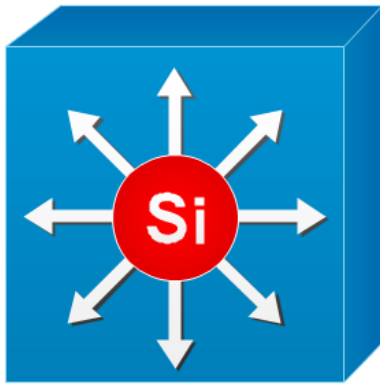
**LSA Flood & LSDB
updated**

Recalculation & restoring

IP STRATEGIC ADDRESSING

Segment Role	Prefix	Subnet Mask	Design Rationale
WAN Point-to-Point	192.168.x.x	255.255.255.252 (/30)	Conserve space, minimize broadcast noise
Inter-Site Transit	10.10.1.0	255.255.255.252 (/30)	L3 Core to Distribution Handoff
Branch LAN (Users)	10.11.20.0	255.255.255.0 (/24)	Scalable local user segmentation
Branch Management	10.11.10.0	255.255.255.0 (/24)	Isolated OOB management access

SWITCHING & SEGMENT ARCHITECTURE



Multilayer switch with Si

Multilayer Switch (MLS)

Inter-VLAN Routing using SVIs

- VLAN 10: Management SVI for device telemetry.
- VLAN 20: End-user data segment with DHCP relay to Core.

Supports routing protocols

- Transit: Routed port FE0/0 (**#no switchport**) isolation on Multi-Layer Switches.

Zero-Trust system: Inbound Extended Access

Control Lists (ACLs)

bound on SVIs to block horizontal lateral movement between different VLANs.

AUTOMATED PROVISIONING

Configuration Deployment Script (**initial_deploy.py**)

Deployment engine utilizing Python and Netmiko for zero-error baseline configuration.

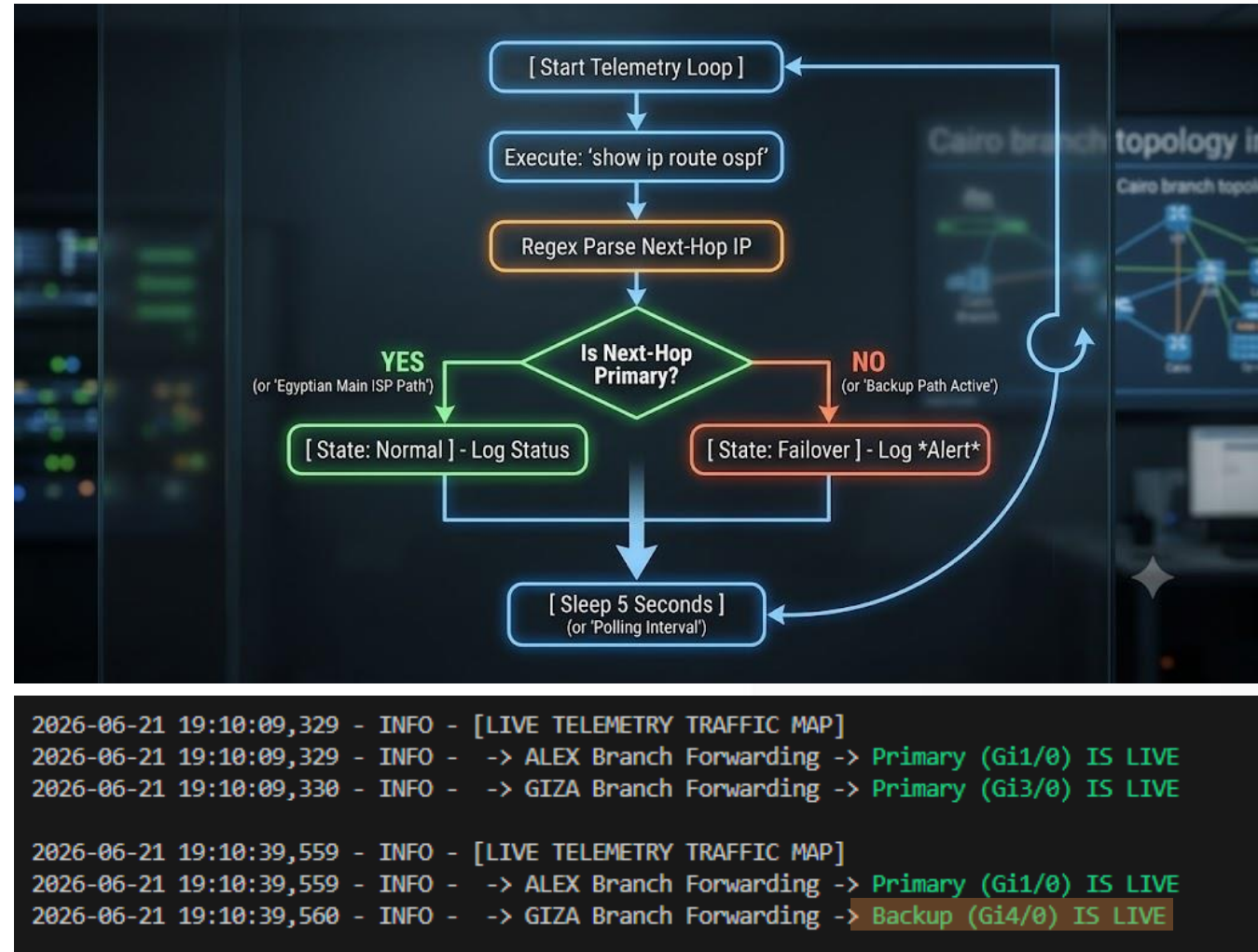
- Separation of Data and Logic
- Programmatic Execution Engine (**Netmiko ConnectHandler**)
- Advanced Dynamic Syntaxes: Dictionary unpacking (****node_param**)
- Defensive Programming Architecture: for fault isolation and mandatory socket teardown. **try/except/finally**

```
def push_config(node_param, config_list):  
    try:  
        print(f"Connecting to port {node_param['port']}...")  
        session = ConnectHandler(**node_param)  
        session.enable()  
        # Clean config pass without tracking prompt status flags  
        session.send_config_set(config_list)  
        session.disconnect()  
        print(f"Port {node_param['port']} configured successfully.\n")  
    except Exception as err:  
        print(f"Error provisioning port {node_param['port']}: {err}")
```

LIVE MONITORING TELEMETRY

Telemetry Logic

- **Background Monitoring Loop:** Runs a continuous **while True** loop with a **time.sleep(30)** delay to check data without overloading the router CPU.
- **OSPF Route Parsing:** Automates continuous text-matching on **show ip route ospf** to find and log active traffic interfaces instantly.
- **Dual-Stream Logging:** Uses the **Python logging** library to send tracking data to the terminal screen and write a timestamped audit trail to **network_status.log**.



Ansible

a shift from raw "Scripting" to true Infrastructure as Code (IaC)

- **Agentless Framework:** no specialized software or agents installed on routers or switches
- **Declarative Execution Model:** handles the underlying execution mechanics natively.
- **Parallel Execution:** using **Fork** concept instead of Sequential Mode device by device



| Questions & Answers ??